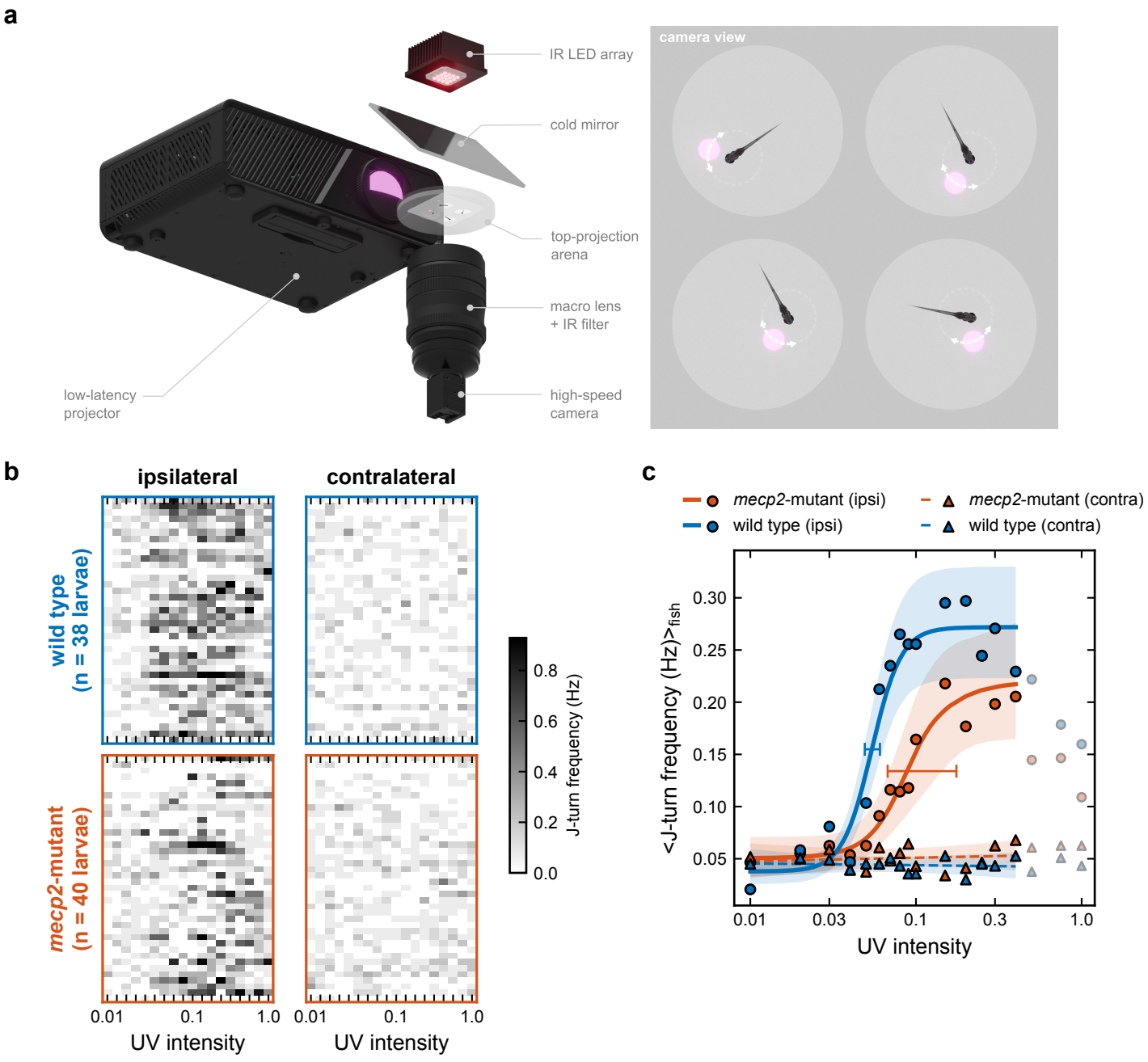

a
IR LED array
cold mirror
top-projection arena
macro lens + IR filter
low-latency projector
high-speed camera
camera view
b
ipsilateral
contralateral
wild type (n = 38 larvae)
mecp2-mutant (n = 40 larvae)
0.01
0.1
1.0
UV intensity
J-turn frequency (Hz)
0.0
0.2
0.4
0.6
0.8
c
mecp2-mutant (ipsi)
wild type (ipsi)
mecp2-mutant (contra)
wild type (contra)
<J-turn frequency (Hz)>fish
0.05
0.10
0.15
0.20
0.25
0.30
0.01
0.03
0.1
0.3
1.0
UV intensity